BC 312 BIOPROCESS TECHNOLOGY (3-0-0)

OBJECTIVE: The aim of this course an in-depth understanding of the key process design concepts relating to the production of biomolecules of industrial importance produced using isolated microbial system.

COURSE CONTENT

UNIT-I

Industrial importance of microorganism. Substrates of fermentation major elements and minor elements

UNIT-II

Substrate for Fermentation Vitamins, Hormones, Trace elements and growth factors, Media design.

UNITIII

Microbial production alcohol and wine, Microbial production of beer, Baker's yeast production, Production of acetone and butanol, Microbial production of glycerol, Microbial production of mushroom

UNIT IV

Microbial production of citric acid and gluconic acid, Microbial production of itaconic acid, gibberellic acid and vinegar, Microbial production of industrial Enzymes

TEXTBOOK AND REFERENCE BOOKS:

1. Industrial Microbiology Prescott and Dunn, CBS Publisher.

2. Comprehensive Biotechnology by Murray and Moo-Young, Academic press.

3. Industrial Microbiology by Casida L.R., New Age International Pvt. Ltd.

4. Microbiology by Pelczar, Chan, and Krieg, TMH.

5. Fermentation Biotechnology, Principles, Processed Products by Ward OP, Open University Press.